

Average of marks for different subjects :

"Think Like Compiler"

- Start
- Declare variables to store the sum of numbers and the average.
- Input 5 numbers.
- Calculate the sum of the 5 numbers.
- Calculate the average using the formula:

$$\text{Average} = \frac{\text{Sum of numbers}}{5}$$

- Display the average.
- End

Enter marks for five subjects:

Subject 1: 85

Subject 2: 90

Subject 3: 78

Subject 4: 88

Subject 5: 92

Total marks: 433.00

Average marks: 86.60

```
import java.util.Scanner;

public class Main{ new *
    public static void main(String[] args) { new *
        Scanner sc=new Scanner(System.in);

        System.out.println("Enter the marks of subject 1:");
        int n1=sc.nextInt();

        System.out.println("Enter the marks of subject 2:");
        int n2=sc.nextInt();

        System.out.println("Enter the marks of subject 3:");
        int n3=sc.nextInt();

        System.out.println("Enter the marks of subject 4:");
        int n4=sc.nextInt();

        System.out.println("Enter the marks of subject 5:");
        int n5=sc.nextInt();

        int total=n1+n2+n3+n4+n5;
        System.out.println("The total marks is "+total);
    }
}
```

```
System.out.println("Enter the marks of subject 5:");
int n5=sc.nextInt();

int total=n1+n2+n3+n4+n5;
System.out.println("The total marks is "+total);

float avg =total/5;
System.out.println("The average marks is "+avg);

}
}
```

Celsius to Fahrenheit :

- Start
- Input: Get the temperature in Celsius from the user.
- Convert: Apply the formula to convert Celsius to Fahrenheit:

$$F = \left(\frac{9}{5} \times C \right) + 32$$

- C is the input temperature in Celsius.
- F is the converted temperature in Fahrenheit.
- Output: Display the converted temperature in Fahrenheit.
- End

Programming with
Real-Life Example:
Celsius to Fahrenheit
Conversion

- Input:** Doctor measures body temperature in Celsius (e.g., 37°C).
- Convert:** Use the formula to convert Celsius to Fahrenheit.
- Output:** 37°C = 98.6°F.

```
import java.util.Scanner;

public class TEMP { new *
    public static void main(String[] args) { new *

        Scanner p1=new Scanner(System.in);
        System.out.println("Enter the temperature in Celsius:");
        double c=p1.nextDouble();
        double F=(1.8*c)+32;//1.8*25+32//45+32=77
        System.out.println("The temperature in Fahrenheit is "+F);
    }
}
```

Temperature conversion :

Conversion Formulas:

- Celsius to Fahrenheit:

$$F = \frac{9}{5} \times C + 32$$

- Fahrenheit to Celsius:

$$C = \frac{5}{9} \times (F - 32)$$

- Celsius to Kelvin:

$$K = C + 273.15$$

- Fahrenheit to Kelvin:

$$K = \frac{5}{9} \times (F - 32) + 273.15$$

Power Calculation

Algorithm:

1. **Start** the program.
2. **Input** the base number and exponent from the user.
3. **Calculate** the power using the formula:

$$\text{Power} = B^E$$

4. **Display** the result to the user.
5. **End** the program.

Real-life Example:

Imagine you want to calculate the amount after compounding with interest where the base is 2 (doubling your initial investment) and you want to see what it becomes after 3 times (exponent 3).

- Base (B) = 2
- Exponent (E) = 3


$$2^3 = 2 \times 2 \times 2 = 8$$

Using the formula, the power would be:

$$\text{Power} = 2^3 = 8$$

```
public class Power { new *  
    public static void main(String[] args) { new *  
  
        int b=2;  
        int e=3;  
  
        int f=b*b*b;  
        System.out.println("2 ki power 3="+f);  
  
        double result=Math.pow(2,3);  
        System.out.println("result="+result);  
    }  
}
```

Simple Interest (SI)

- **Start** the program.
- **Input** the principal amount, rate of interest, and time period from the user.
- **Calculate** the Simple Interest using the formula

$$SI = \frac{P \times R \times T}{100}$$

- **Display** the result (Simple Interest) to the user.
- **End** the program.

Imagine you invest \$1,000 in a savings account that offers an annual interest rate of 5% for 3 years. The Simple Interest (SI) can be calculated as:

- Principal (P) = \$1000
- Rate (R) = 5% per year
- Time (T) = 3 years

Using the formula, the Simple Interest will be:

$$SI = \frac{1000 \times 5 \times 3}{100} = 150$$

Thus, the interest earned over 3 years would be **\$150.**

```
import java.util.Scanner;

public class SI { new *
    public static void main(String[] args) { new *
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the P:");
        double P=sc.nextDouble();

        System.out.println("Enter the R:");
        double R=sc.nextDouble();

        System.out.println("Enter the T:");
        double T=sc.nextDouble();

        double SI=(P*R*T)/100;
        System.out.println("The Simple Inter:"+SI);

    }
}
```

Compound Interest (CI)

Algorithm:

1. Start the program.
2. Input the principal amount, rate of interest, and time period from the user.
3. Calculate the final amount (A) using the formula:

$$A = P \times \left(1 + \frac{R}{100}\right)^T$$

4. Calculate the Compound Interest (CI) by subtracting the principal from the final amount:

$$CI = A - P$$

5. Display the result (Compound Interest) to the user.
6. End the program.

Real-life Example:

Imagine you invest \$1,000 in a savings account that compounds annually at a rate of 5% for 3 years.

- Principal (P) = \$1000
- Rate (R) = 5%
- Time (T) = 3 years

Using the formula, the final amount (A) would be:

$$A = 1000 \times \left(1 + \frac{5}{100}\right)^3 = 1000 \times 1.157625 = 1157.625$$

Therefore, the Compound Interest (CI) is:

$$CI = 1157.625 - 1000 = 157.625$$

Method 3: Using XOR (Bitwise Assignment Operator)

```
public class SwapUsingXOR {  
    public static void main(String[] args) {  
        int a = 5, b = 10;  
  
        System.out.println("Before Swap: a = " + a + ", b = " + b);  
  
        a ^= b; // a = a ^ b  
        b ^= a; // b = b ^ a  
        a ^= b; // a = a ^ b  
  
        System.out.println("After Swap: a = " + a + ", b = " + b);  
    }  
}
```

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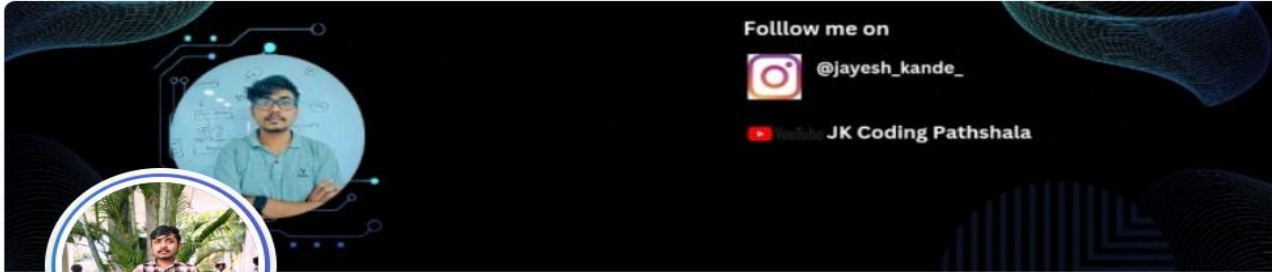
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